# NAAN MUDHALVAN PROJECT

## TITLE NAME: FARM MANAGEMENT SYSTEM (FMS)

## PROJECT REPORT

## DEVELOPED BY

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**KRISHNASAMY COLLEGE OF SCIENCE, ARTS AND MANAGEMENT FOR WOMEN**

## USER STORY

In the Farm Management System (FMS), farmers, farm managers, and administrators use a centralized Salesforce platform to manage farm, crop, farmer, and buyer information. The system allows users to record and update agricultural data easily with proper validation rules to avoid incorrect entries.

It also provides reports and dashboards to analyze crop production, farm performance, and sales activities. The platform ensures secure data storage with role-based access so that sensitive information is protected. This system reduces manual paperwork, improves accuracy, and supports better farm management decisions.

# 1. INTRODUCTION

### 1.1 Project Overview

The **Farm Management System (FMS)** is a Salesforce-based application developed to manage agricultural data digitally. It helps store and organize information related to farms, crops, farmers, and buyers.

Using Salesforce features such as **custom objects, relationships, reports, and dashboards**, the system helps users monitor farm activities and analyze performance in real time.

### 1.2 Purpose

The main purpose of this project is to:

• Digitize farm data management
• Improve data accuracy
• Reduce manual record keeping
• Generate real-time reports and dashboards
• Help farmers and managers make better decisions

# 2. IDEATION PHASE

### 2.1 Problem Statement

| User | Problem | Reason | Result |
|---|---|---|---|
| Farmer | Managing farm records | Records stored manually | Data errors |
| Farm Manager | Monitoring production | Data scattered in files | Delay in reports |
| Buyer | Access crop information | No centralized system | Uncertain information |
| Admin | Managing system security | Manual access control | Security risks |

### 2.2 Empathy Map

**Target Users**

• Farmers
• Farm Managers
• Buyers
• System Administrators

**Analysis**

Think & Feel – Need accurate farm data
Hear – Advice from agricultural experts
See – Paper records and spreadsheets
Say & Do – Manually update records
Pain – Data loss and errors
Gain – Automated system with fast reporting

### 2.3 Brainstorming

Possible solutions discussed:

• Manual farm records
• Excel-based system
• Web application
• Mobile application
• Salesforce-based management system

**Selected Idea**

Salesforce-based **Farm Management System**

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey

1 Login to system
2 Open Farm Management App
3 Enter farm and crop data
4 Register farmers and buyers
5 Generate reports
6 Analyze dashboard results
7 Make decisions based on data

### 3.2 Solution Requirements

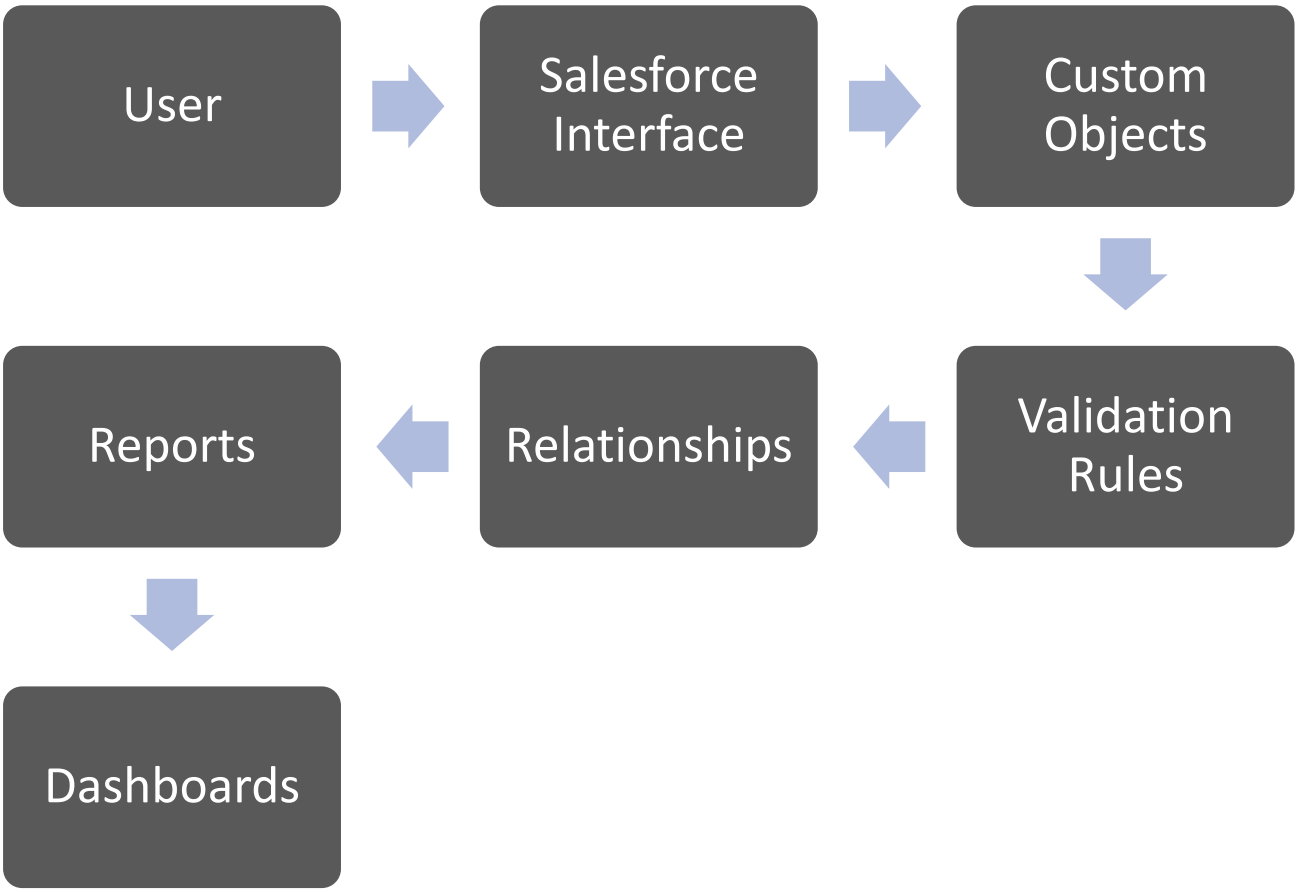
**Functional Requirements**

FR1 User Login and Authentication
FR2 Manage Farms, Crops, Farmers, Buyers
FR3 Data entry with validation
FR4 Relationship between objects
FR5 Generate Reports
FR6 Dashboard Visualization
FR7 User Role Management

**Non Functional Requirements**

| Requirement | Description |
|---|---|
| Usability | Easy interface using Lightning UI |
| Security | Data protection using profiles and roles |
| Reliability | Cloud storage prevents data loss |
| Performance | Fast report generation |
| Availability | 24/7 cloud access |
| Scalability | Supports more users and farms |

### 3.3 Data Flow Diagram




### 3.4 Technology Stack

| Layer | Technology |
|---|---|
| **Frontend** | Salesforce Lightning UI |
| **Backend** | Salesforce Platform |
| **Database** | Salesforce Cloud |
| **Automation** | Validation Rules |
| **Analytics** | Reports and Dashboards |

# 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

Traditional farming records are usually maintained in paper registers, which can lead to data loss and errors.

The **Farm Management System** solves this problem by storing all data in Salesforce cloud storage. Reports and dashboards provide quick insights into farm productivity and sales.

### 4.2 Proposed Solution

Problem: Manual farm record management.

Solution:
A Salesforce-based system that stores farm, crop, farmer, and buyer data in one centralized platform.

Benefits:

• Accurate data storage
• Automated validations
• Faster report generation
• Better decision making

### 4.3 Solution Architecture

System Architecture Layers

1 Presentation Layer – Salesforce Lightning UI
2 Application Layer – Validation Rules and Automation
3 Data Layer – Custom Objects
4 Analytics Layer – Reports and Dashboards

# 5. PROJECT PLANNING

### 5.1 Project Steps

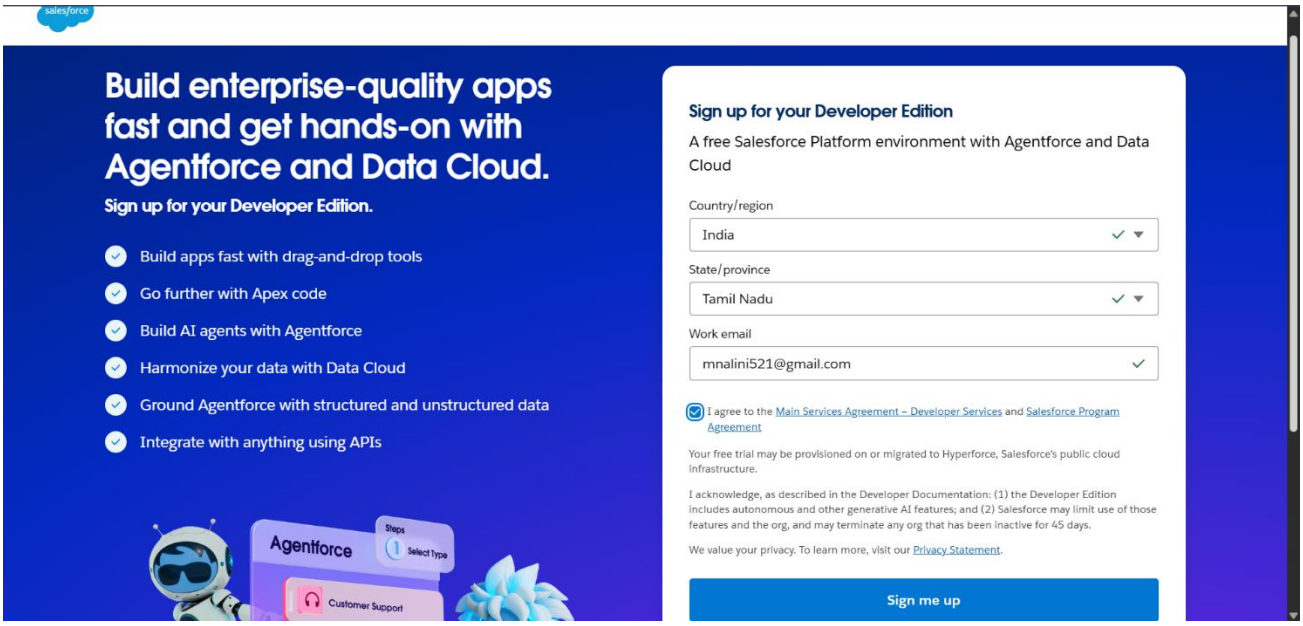
• Create Salesforce Developer Account
• Create Custom Objects
• Create Fields and Relationships
• Configure Tabs and Lightning App
• Create Profiles and Roles
• Create Users
• Configure Sharing Rules
• Create Reports and Dashboards
• Test the application

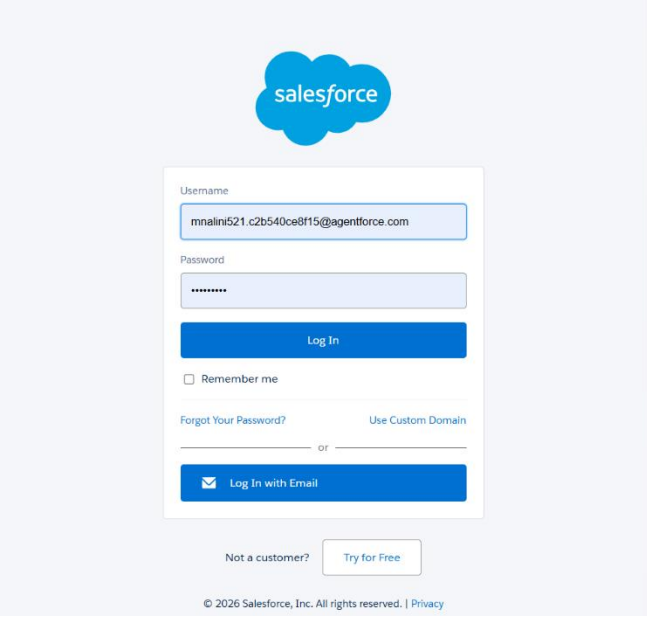
# 6. PROJECT DEVELOPMENT

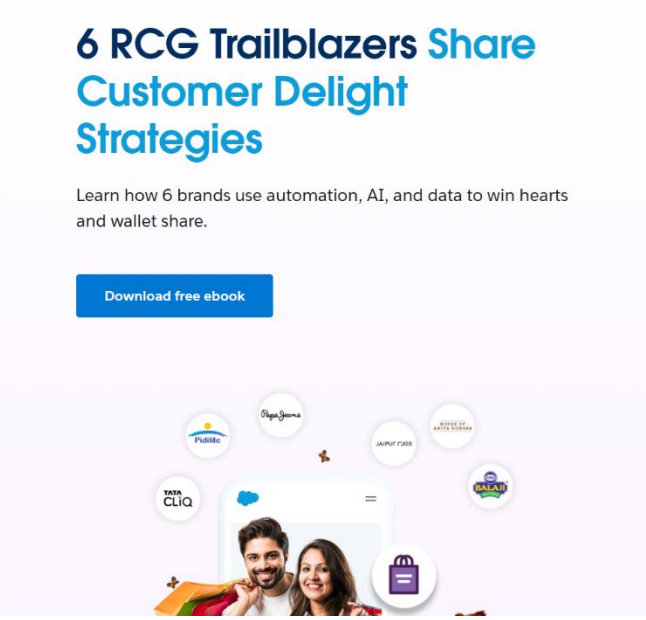
## What is Salesforce

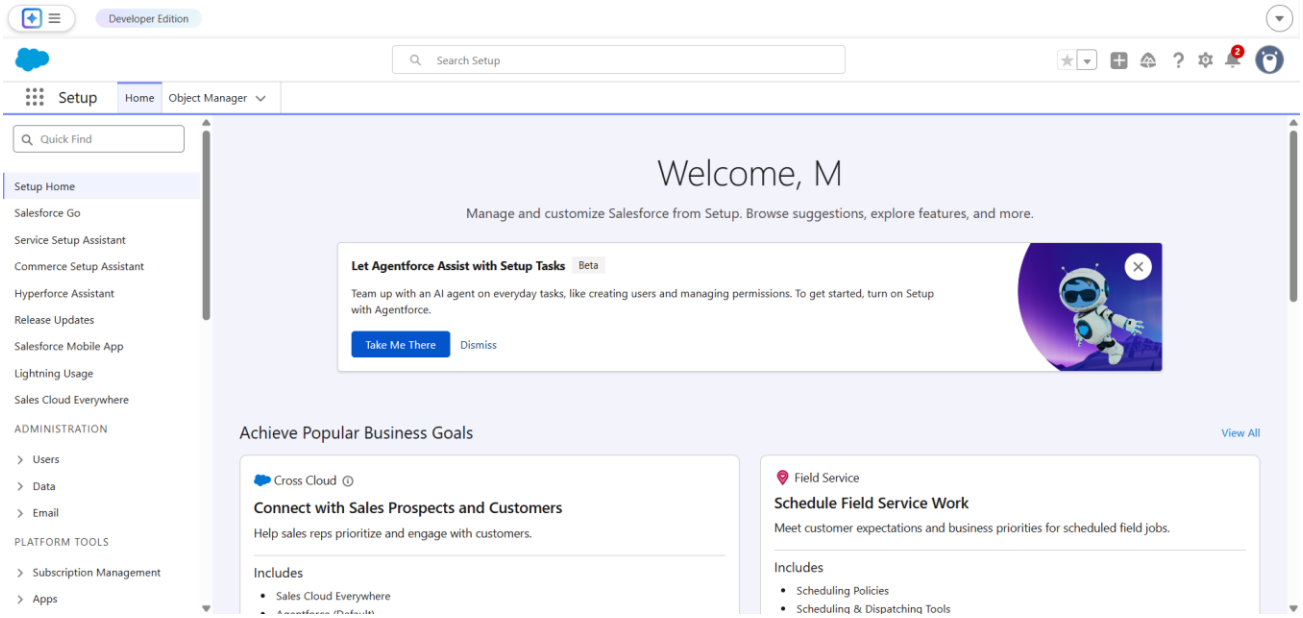
Salesforce is a cloud-based CRM platform that helps organizations manage data, automate processes, and analyze business performance.

It allows users to create applications using objects, fields, reports, and dashboards without complex coding.



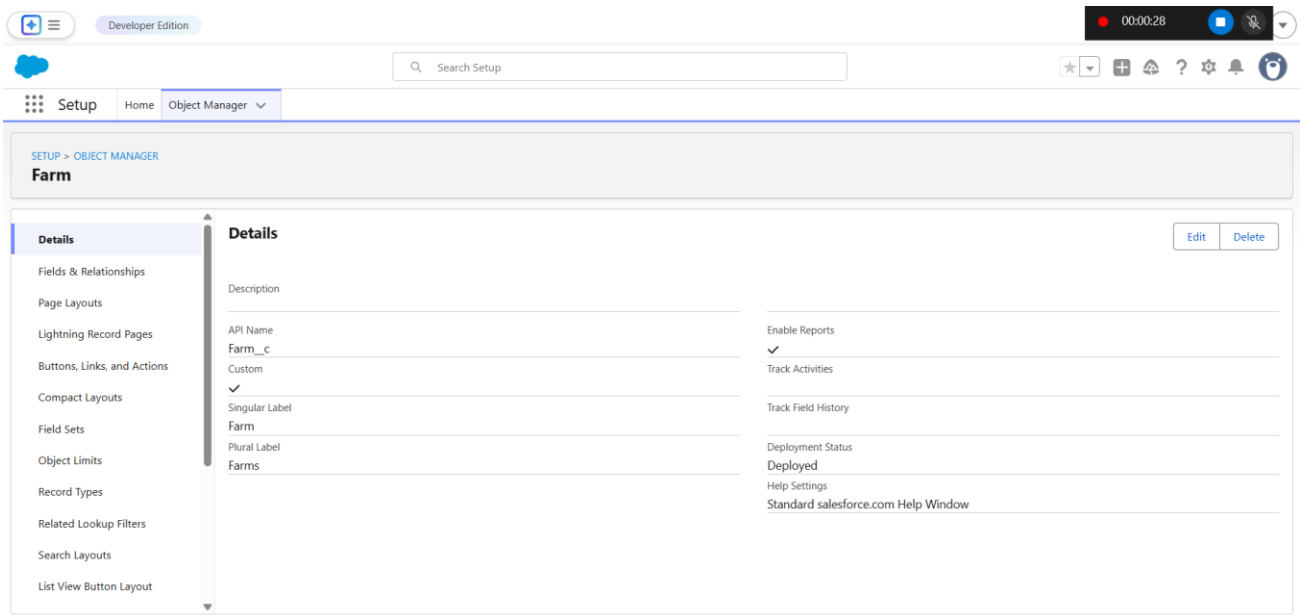






## Custom Objects Used

In this project four custom objects were created:

- Farm
- Crop
- Farmer
- Buyer



Developer Edition

00:04:02

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

## Crop

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

### Details

Edit | Delete

Description

API Name
Crop__c

Custom
✓

Singular Label
Crop

Plural Label
Crops

Enable Reports
✓

Track Activities

Track Field History

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window

Developer Edition

00:06:16

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

## Farmer

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

### Details

Edit | Delete

Description

API Name
Farmer__c

Custom
✓

Singular Label
Farmer

Plural Label
Farmers

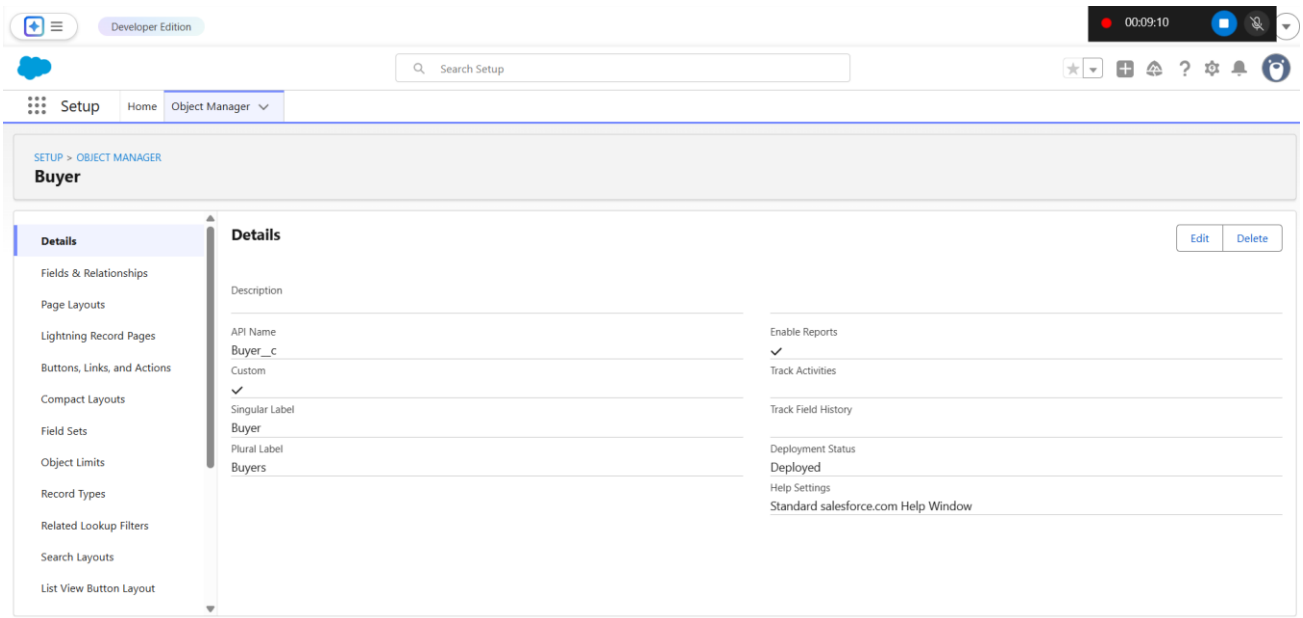
Enable Reports
✓

Track Activities

Track Field History

Deployment Status
Deployed

Help Settings
Standard salesforce.com Help Window



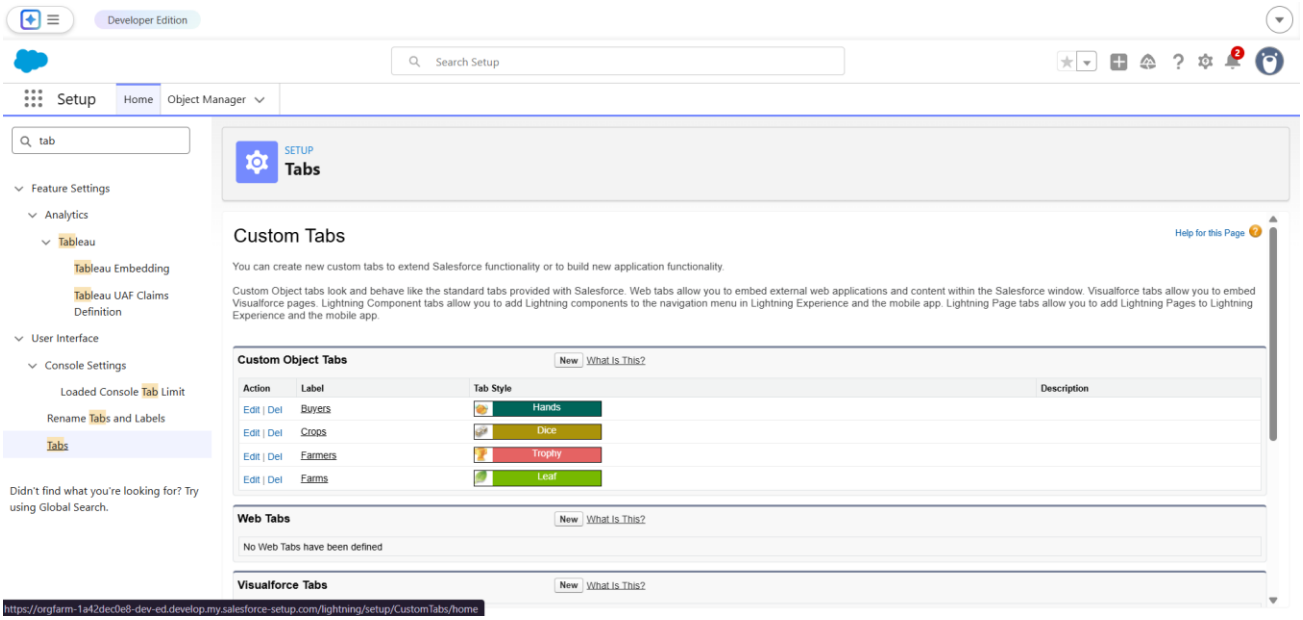

(Screenshot – Object Creation)

## Tabs

Tabs were created for each object so users can easily access the data.

Tabs created:

- Farm
- Crop
- Farmer
- Buyer



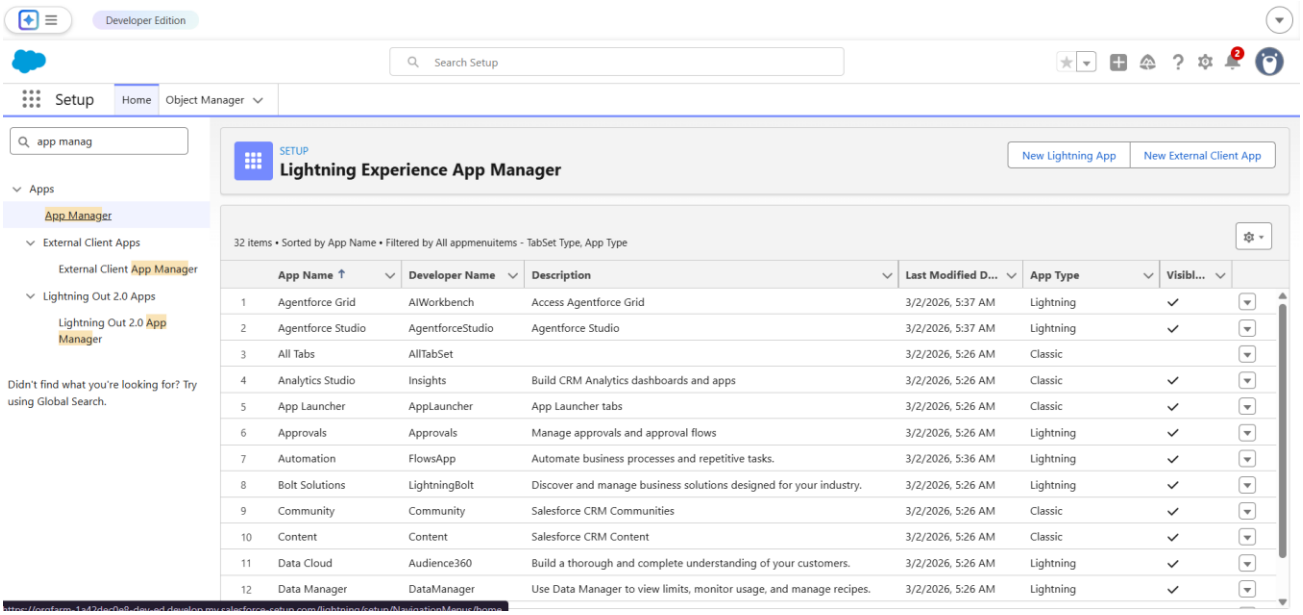

(Screenshot – Tabs)

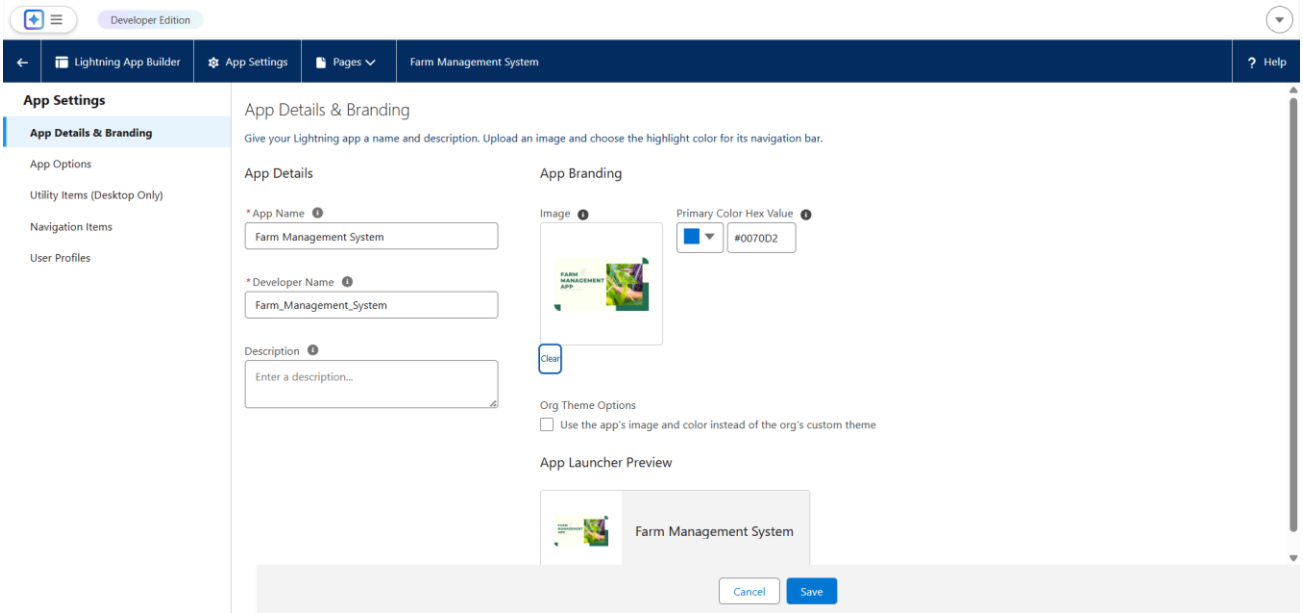
# LIGHTNING APP

A **Farm Management System Lightning App** was created using App Manager.

The following tabs were included:

- Farm
- Crop
- Farmer
- Buyer
- Reports
- Dashboards




(Screenshot – Lightning App)

## FIELDS AND RELATIONSHIPS

Fields were created for each object to store information.

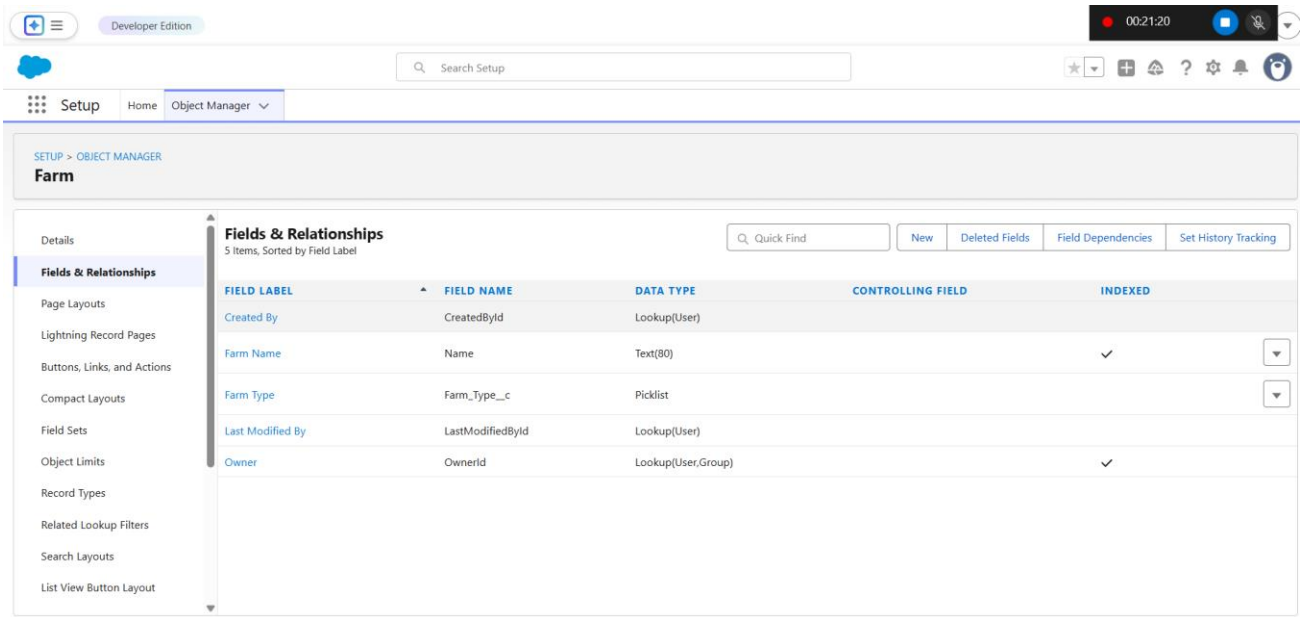
Examples:

Farm Object
• Farm Type
• Soil Quality
• Irrigation Type
• Location
• Size

Crop Object
• Crop Name
• Planting Date
• Harvest Date
• Yield

Relationships were created between objects such as **Farmer and Farm**, and **Farm and Crop**.

(Screenshot – Fields and Relationships)



SETUP > OBJECT MANAGER

## Crop

- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

### Fields & Relationships

12 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

| FIELD LABEL | FIELD NAME | DATA TYPE | CONTROLLING FIELD | INDEXED |
|---|---|---|---|---|
| Actual Yield | Actual_Yield__c | Currency(18, 0) | | |
| Created By | CreatedById | Lookup(User) | | |
| Crop Name | Crop_Name__c | Text(80) | | |
| Crop Name | Name | Text(80) | | ✓ |
| Expected Yield per Acre | Expected_Yield_per_Acre__c | Formula (Currency) | | |
| Farm | Farm__c | Lookup(Farm) | | ✓ |
| Harvest Date | Harvest_Date__c | Date | | |
| Last Modified By | LastModifiedById | Lookup(User) | | |

Developer Edition | 00:11:39

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

## Farmer

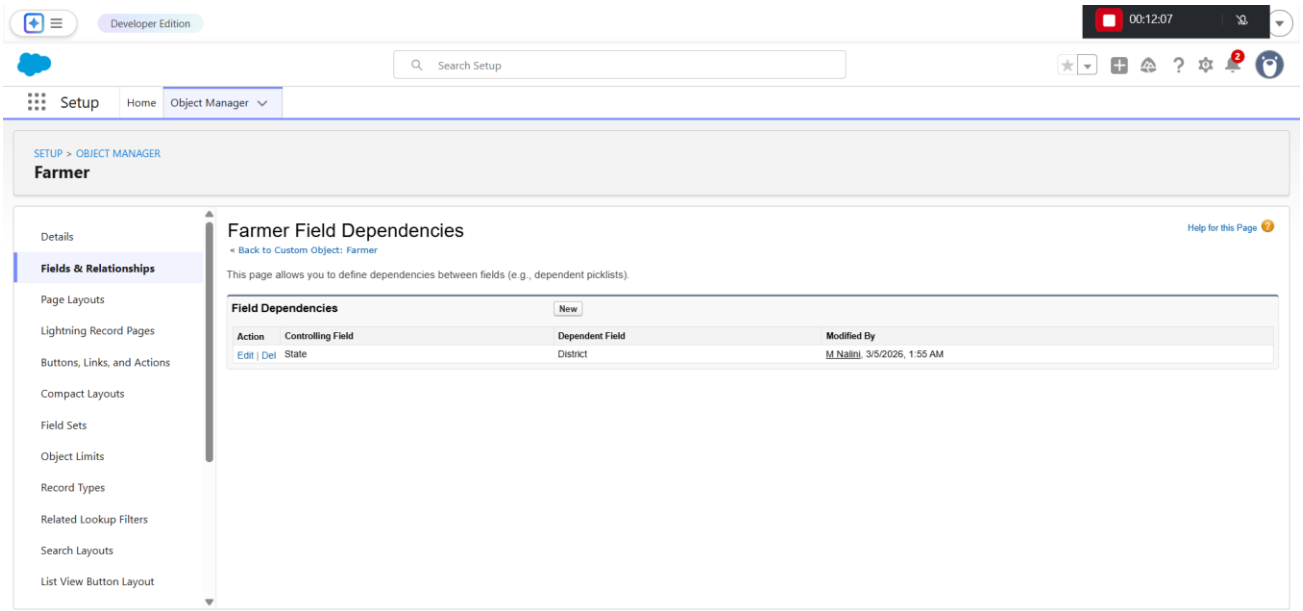
- Details
- **Fields & Relationships**
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Search Layouts
- List View Button Layout

### Fields & Relationships

8 Items, Sorted by Field Label

Quick Find | New | Deleted Fields | Field Dependencies | Set History Tracking

| FIELD LABEL | FIELD NAME | DATA TYPE | CONTROLLING FIELD | INDEXED |
|---|---|---|---|---|
| Contact Number | Contact_Number__c | Phone | | |
| Created By | CreatedById | Lookup(User) | | |
| District | District__c | Picklist | State | |
| Farmer Name | Name | Text(80) | | ✓ |
| Last Modified By | LastModifiedById | Lookup(User) | | |
| Owner | OwnerId | Lookup(User,Group) | | ✓ |
| State | State__c | Picklist | | |
| Village | Village__c | Picklist | | |


Developer Edition
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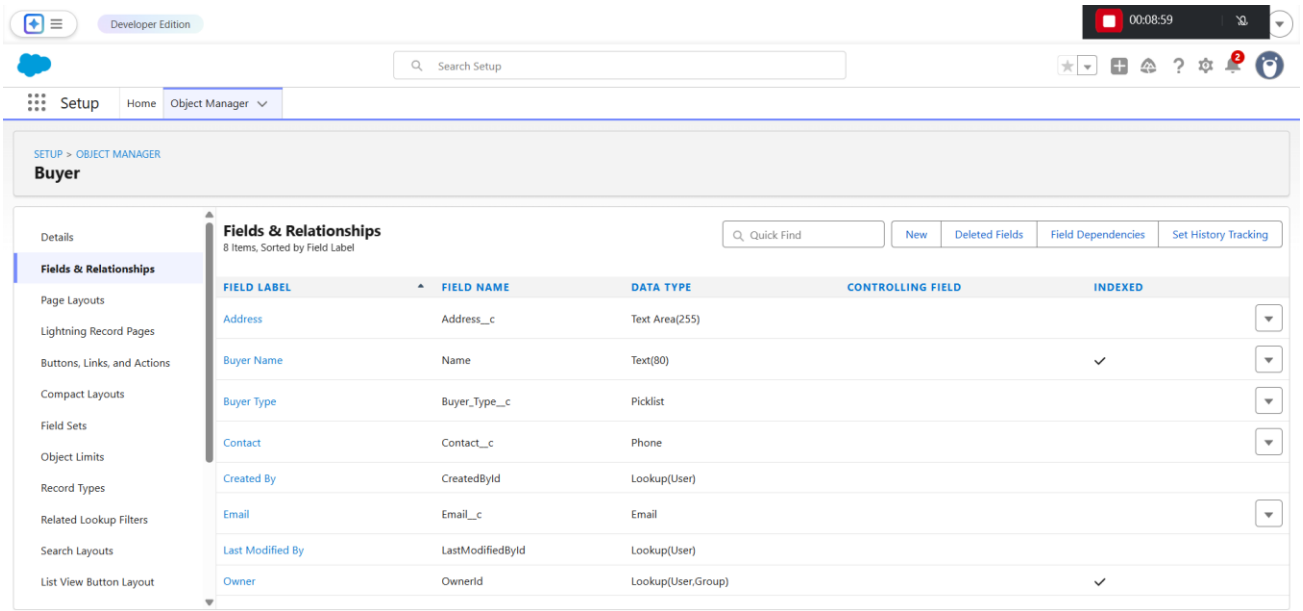
Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Farmer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Farmer Field Dependencies
Help for this Page
« Back to Custom Object: Farmer
This page allows you to define dependencies between fields (e.g., dependent picklists).
Field Dependencies
New
Action
Controlling Field
Dependent Field
Modified By
Edit | Del
State
District
M Nalini, 3/5/2026, 1:55 AM


Developer Edition
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Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Buyer
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
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Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout
Fields & Relationships
8 Items, Sorted by Field Label
Quick Find
New
Deleted Fields
Field Dependencies
Set History Tracking
FIELD LABEL
FIELD NAME
DATA TYPE
CONTROLLING FIELD
INDEXED
Address
Address__c
Text Area(255)
Buyer Name
Name
Text(80)
Buyer Type
Buyer_Type__c
Picklist
Contact
Contact__c
Phone
Created By
CreatedById
Lookup(User)
Email
Email__c
Email
Last Modified By
LastModifiedById
Lookup(User)
Owner
OwnerId
Lookup(User,Group)

# PROFILES AND ROLES

Profiles and roles were created to control user access.

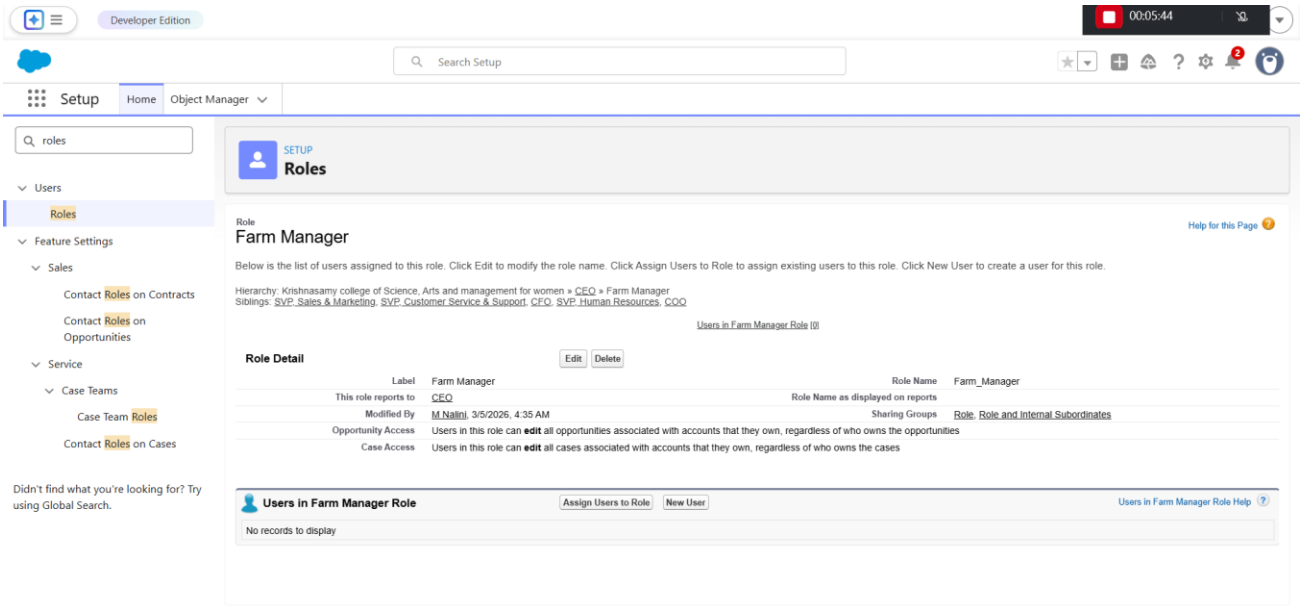
Profiles Created:

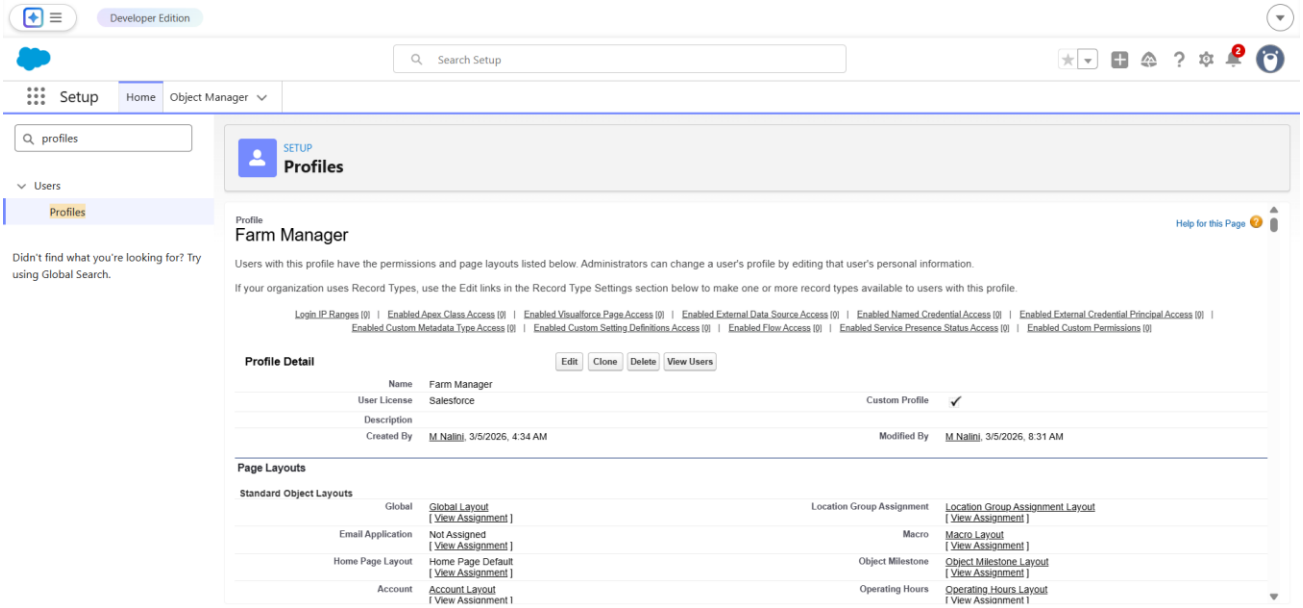
- Farm Manager
- Sales Department

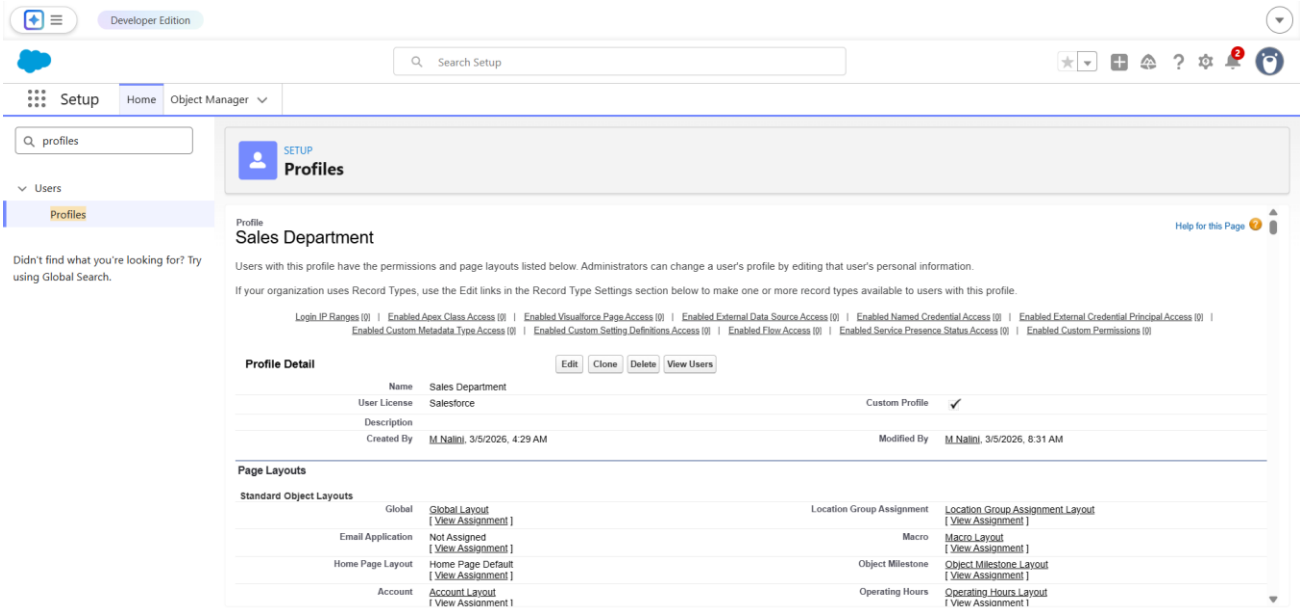
Role Created:

- Farm Manager

(Screenshot – Profiles and Roles)





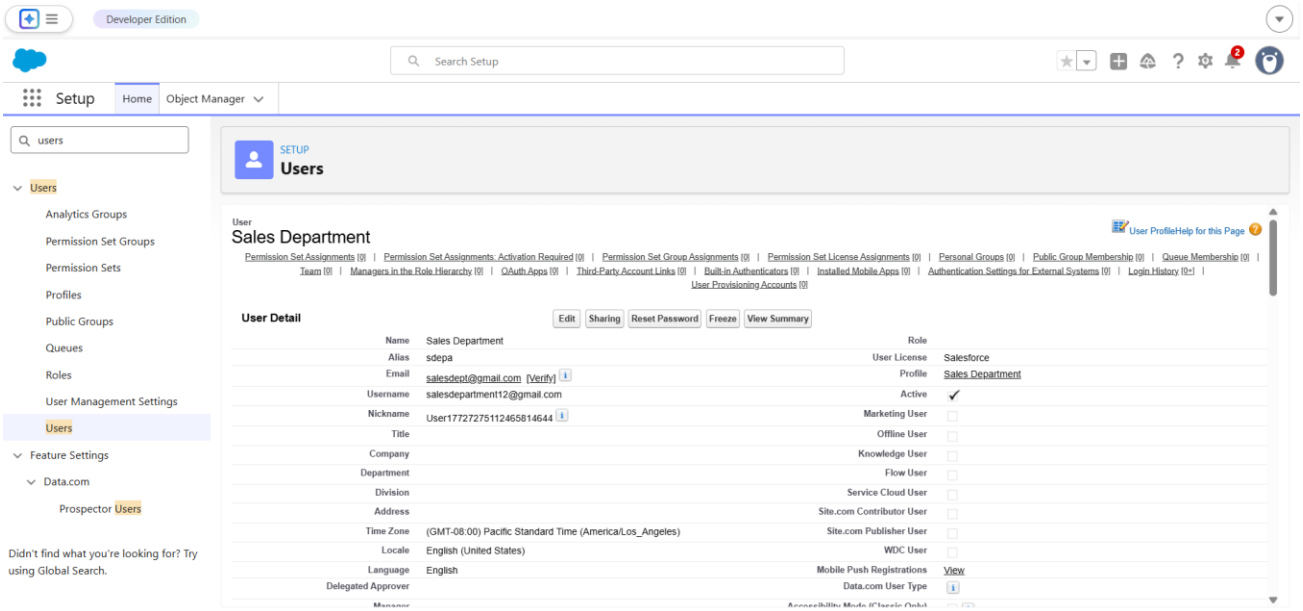

# USERS

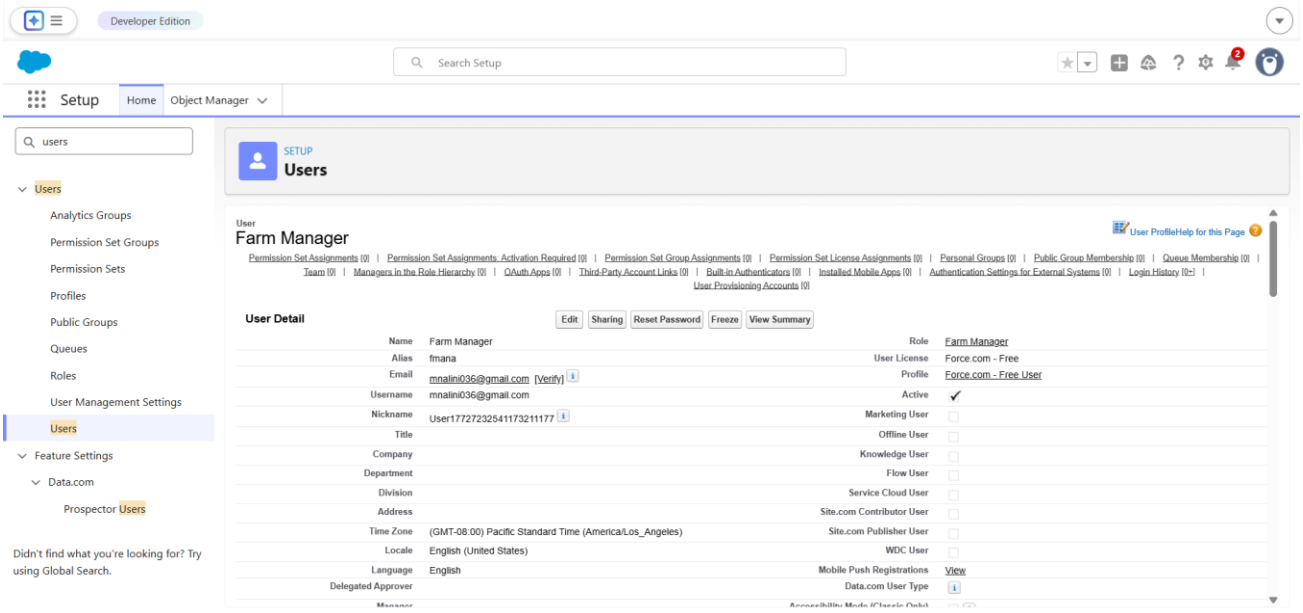
Users were created to access the system.

Example:

Farm Manager User

(Screenshot – User Creation)





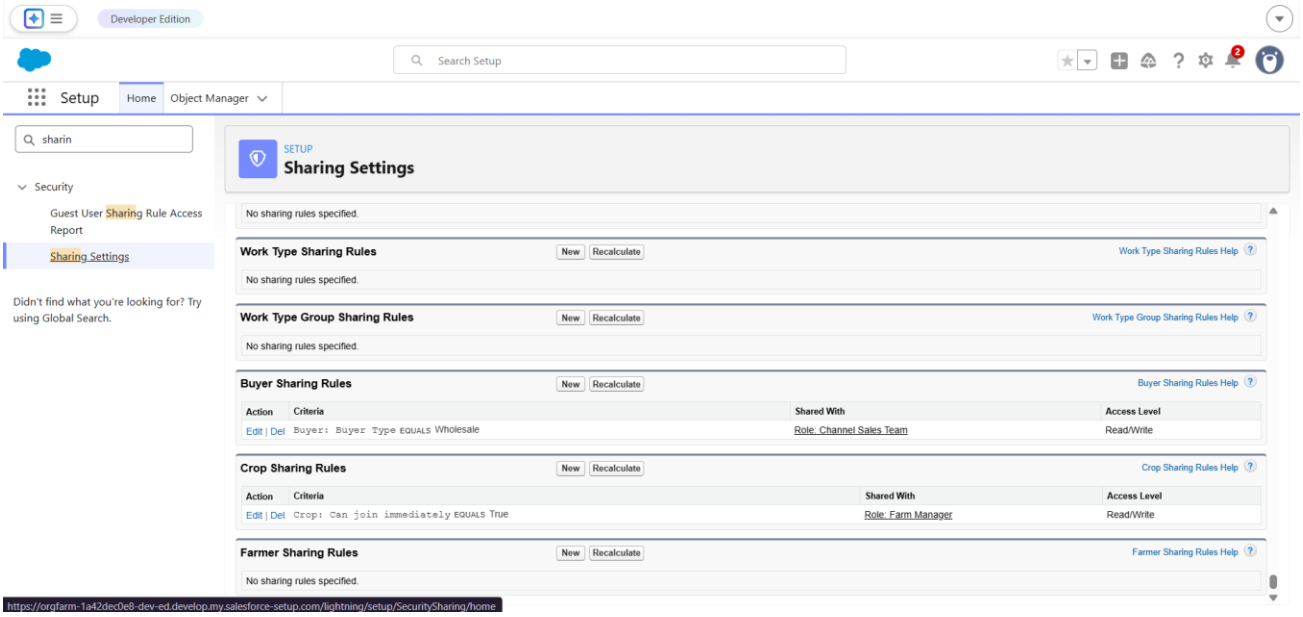

# SHARING RULES

Sharing rules were created to allow data access between users.

Example:

Crop records shared with Farm Manager.

(Screenshot – Sharing Rules)



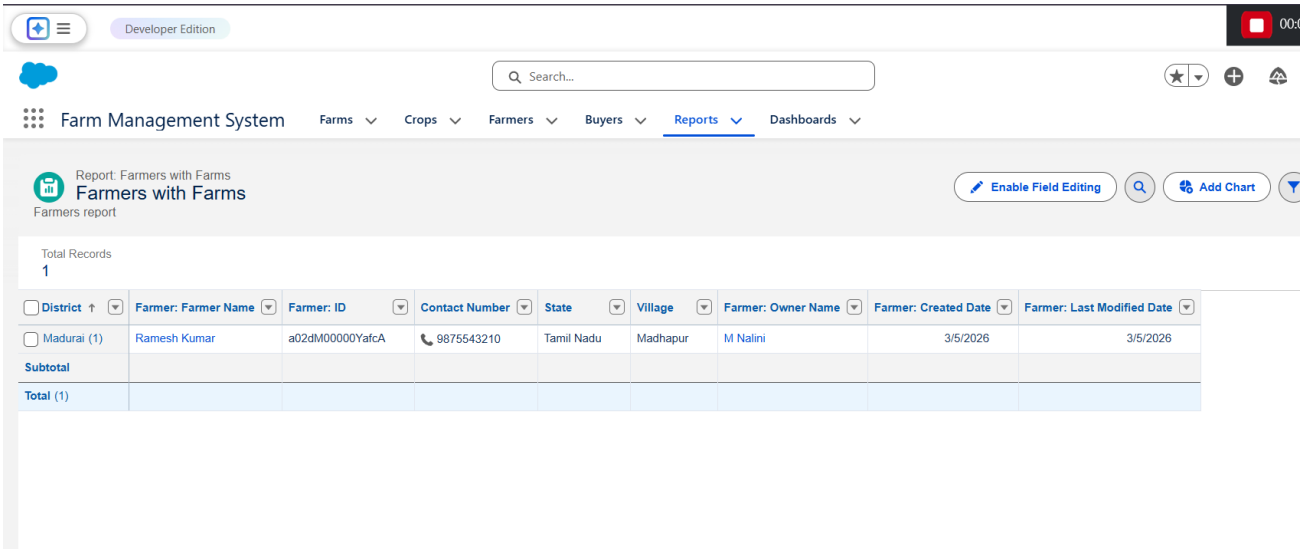
# REPORTS

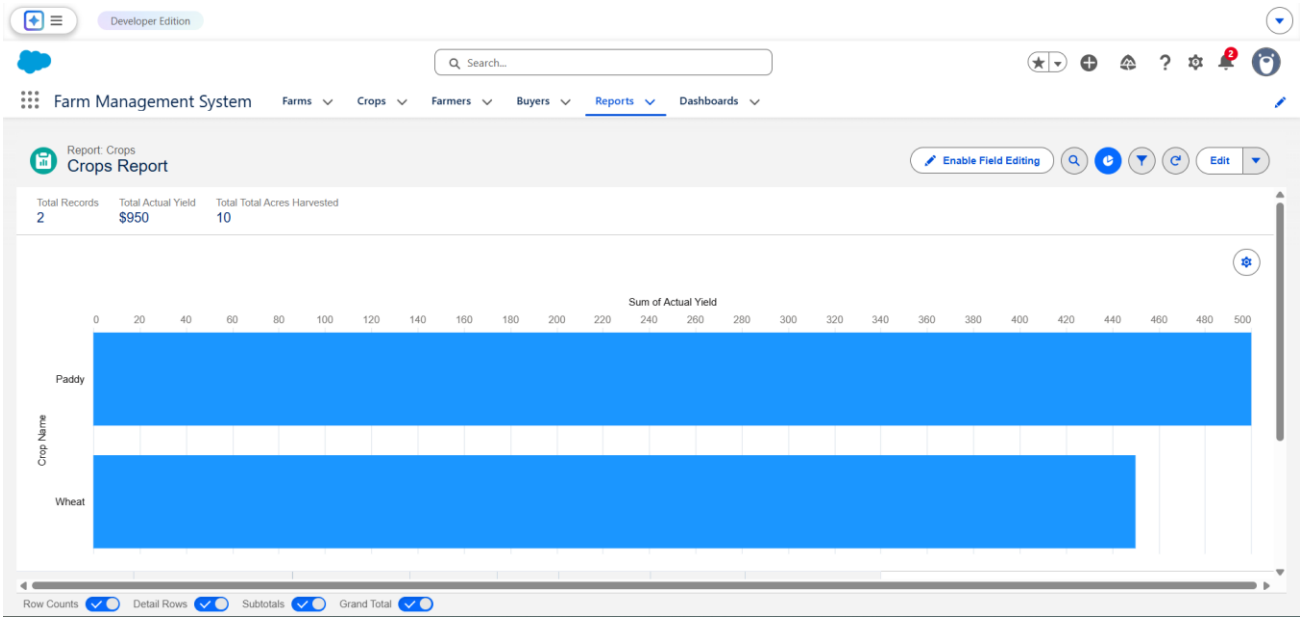
Reports help analyze stored data.

Example Reports:

- Farmers with Farms
- Crops with Buyers

(Screenshot – Reports)



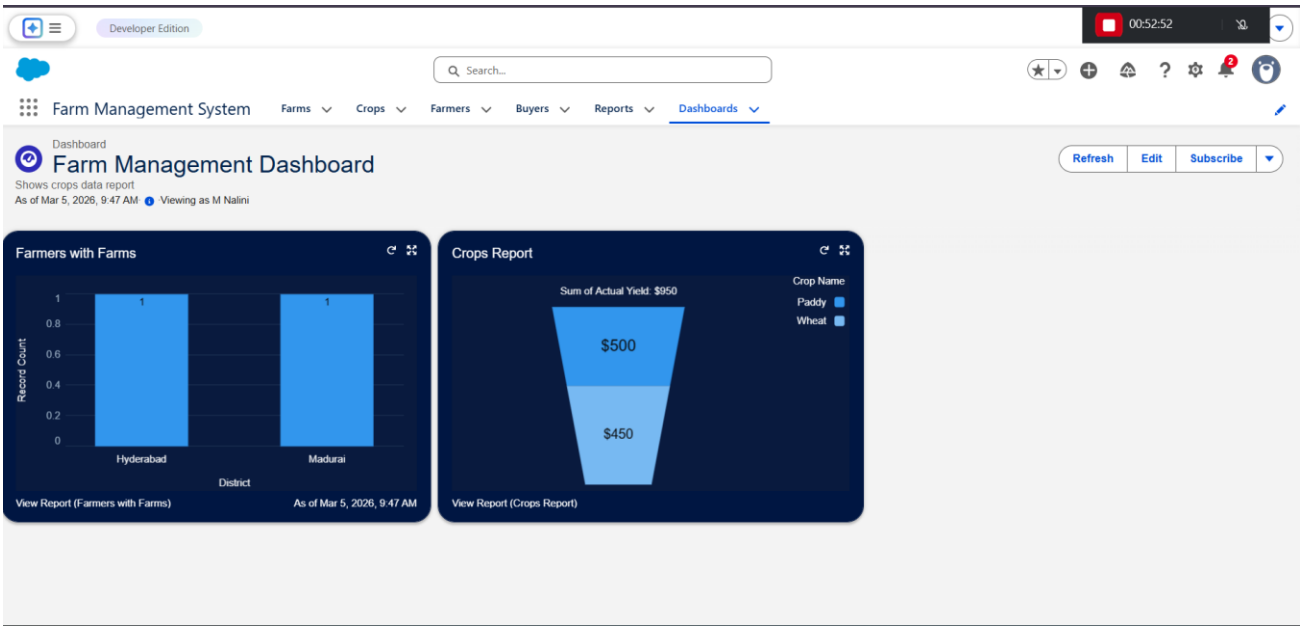
## DASHBOARDS

Dashboards display summarized data using charts and graphs.

Example Dashboard:

• Farmers with Farms

(Screenshot – Dashboard)

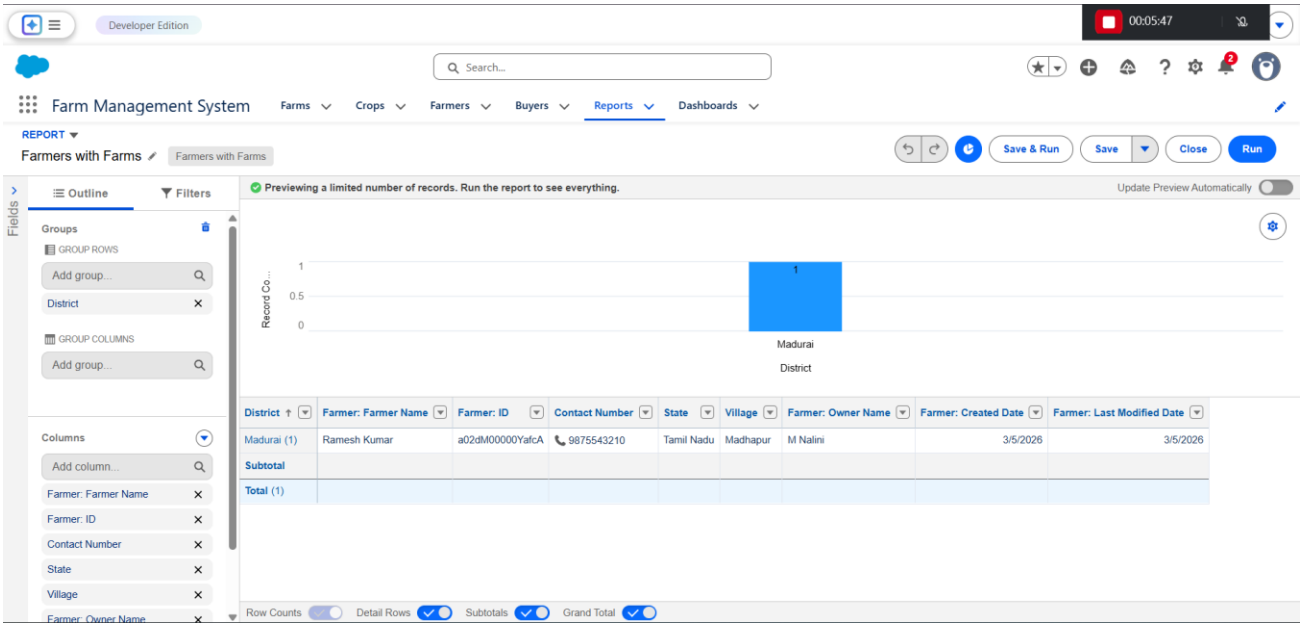


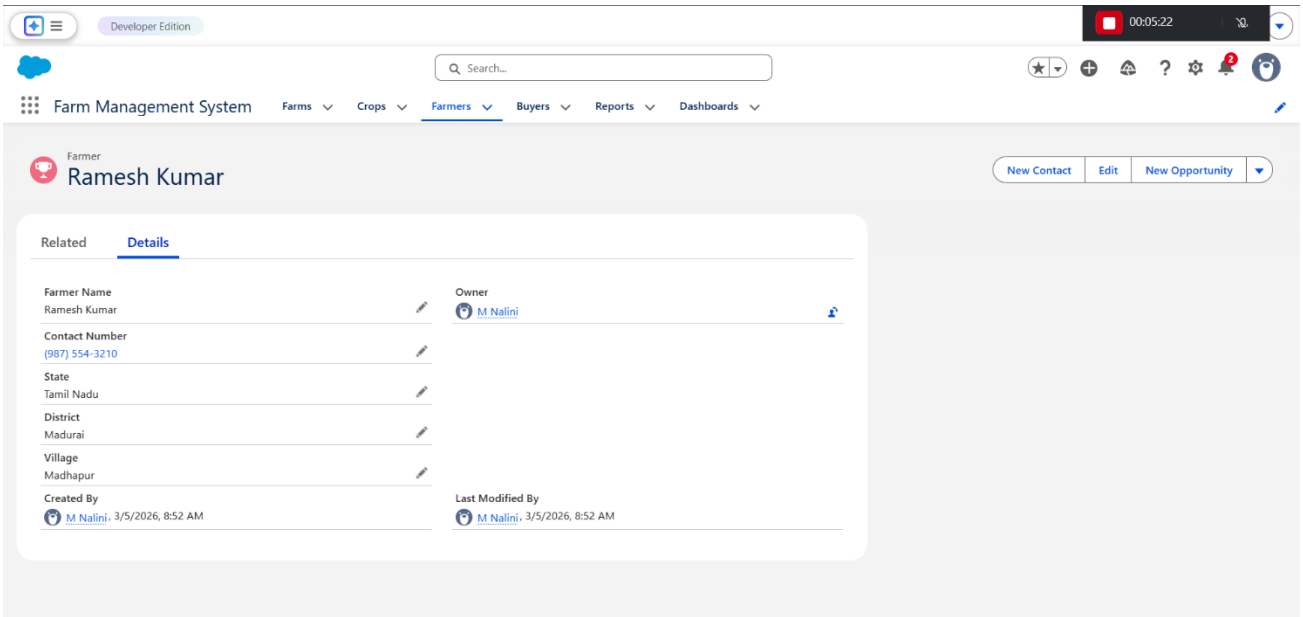

## 7. TESTING

During testing the following features were verified:

• Record creation
• Field validations
• Object relationships
• Profile and role access
• Sharing rules
• Report generation
• Dashboard display

Screenshots were captured to verify system functionality.


Developer Edition
00:05:47
Search...
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
REPORT
Farmers with Farms
Farmers with Farms
Save & Run
Save
Close
Run
Fields
Outline
Filters
Previewing a limited number of records. Run the report to see everything.
Update Preview Automatically
Groups
GROUP ROWS
Add group...
District
GROUP COLUMNS
Add group...
Record Co...
1
0.5
0
1
Madurai
District
Columns
Add column...
Farmer: Farmer Name
Farmer: ID
Contact Number
State
Village
Farmer: Owner Name
District
Farmer: Farmer Name
Farmer: ID
Contact Number
State
Village
Farmer: Owner Name
Farmer: Created Date
Farmer: Last Modified Date
Madurai (1)
Ramesh Kumar
a02dM00000YafcA
9875543210
Tamil Nadu
Madhapur
M Nalini
3/5/2026
3/5/2026
Subtotal
Total (1)
Row Counts
Detail Rows
Subtotals
Grand Total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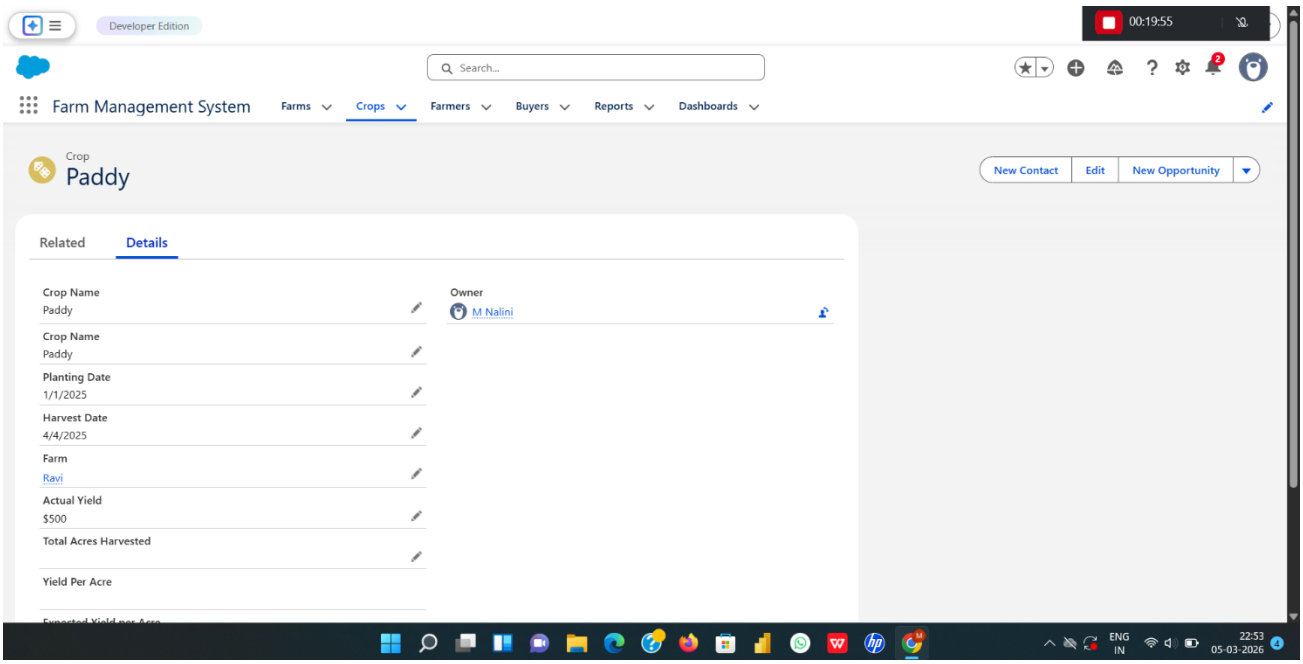
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Farmer
Ramesh Kumar
New Contact
Edit
New Opportunity
Related
Details
Farmer Name
Ramesh Kumar
Owner
M Nalini
Contact Number
(987) 554-3210
State
Tamil Nadu
District
Madurai
Village
Madhapur
Created By
M Nalini, 3/5/2026, 8:52 AM
Last Modified By
M Nalini, 3/5/2026, 8:52 AM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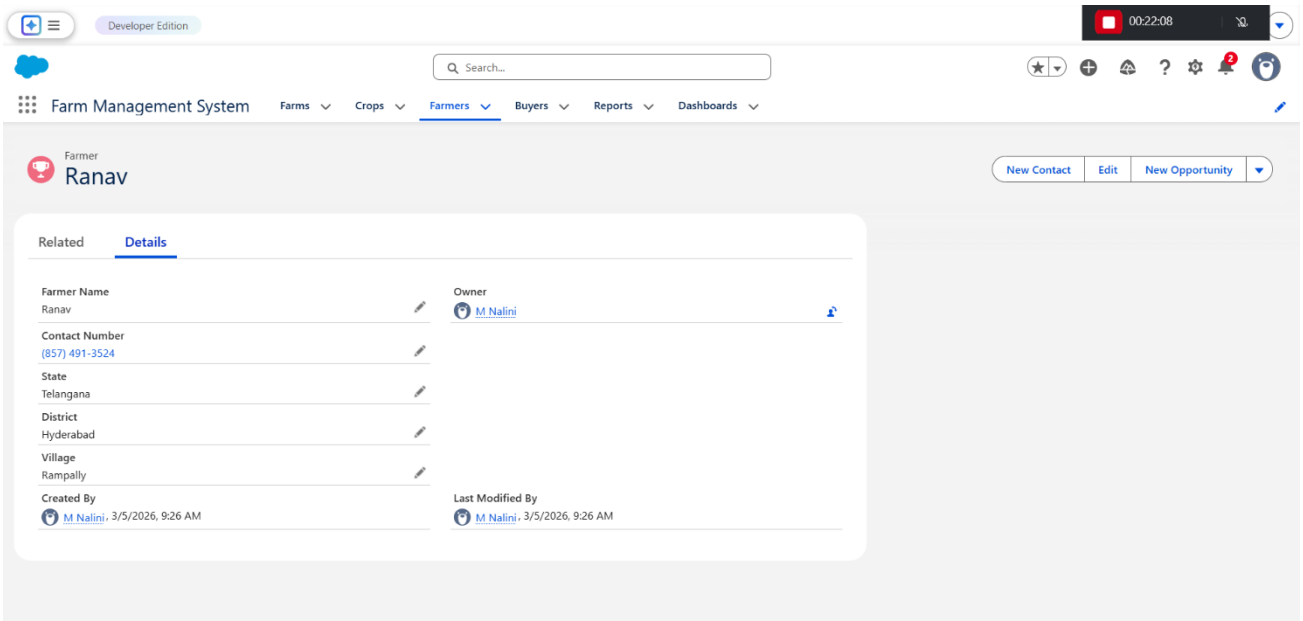
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Crop
Paddy
New Contact
Edit
New Opportunity
Related
Details
Crop Name
Paddy
Owner
M Nalini
Crop Name
Paddy
Planting Date
1/1/2025
Harvest Date
4/4/2025
Farm
Ravi
Actual Yield
$500
Total Acres Harvested
Yield Per Acre
ENG
IN
22:53
05-03-2026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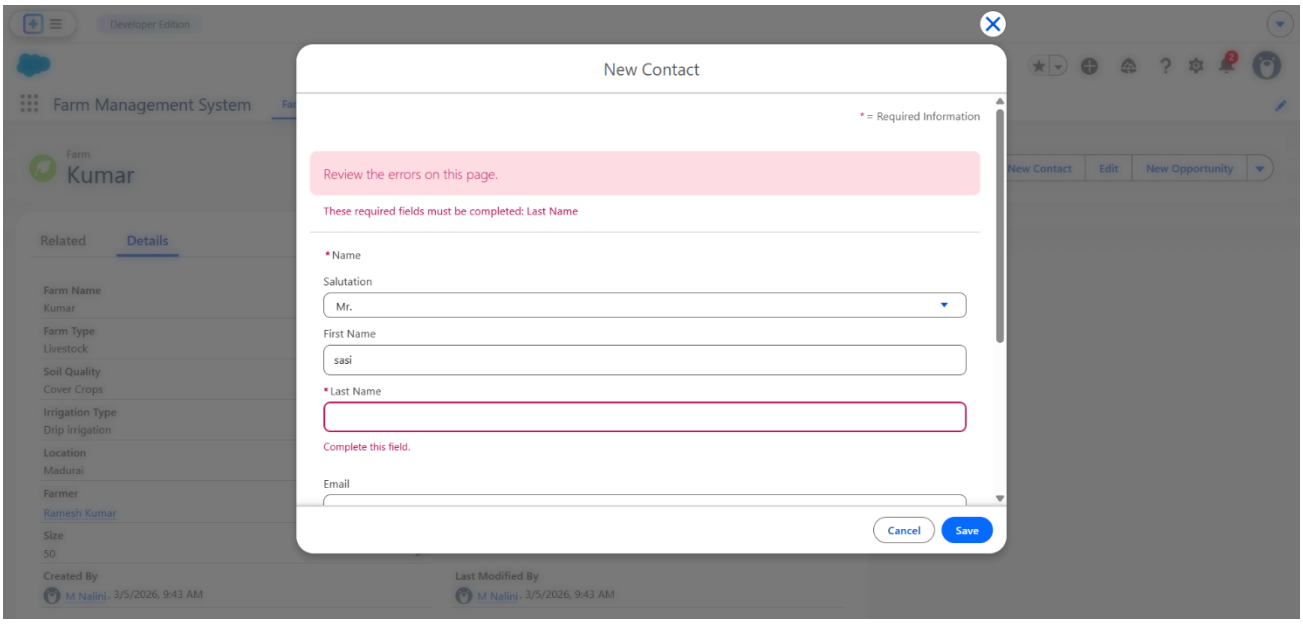
Farm Management System
Farms
Crops
Farmers
Buyers
Reports
Dashboards
Farmer
Ranav
New Contact
Edit
New Opportunity
Related
Details
Farmer Name
Ranav
Owner
M Nalini
Contact Number
(857) 491-3524
State
Telangana
District
Hyderabad
Village
Rampally
Created By
M Nalini, 3/5/2026, 9:26 AM
Last Modified By
M Nalini, 3/5/2026, 9:26 AM




## 8. ADVANTAGES & DISADVANTAGES

### Advantages

• Centralized data storage
• Reduced paperwork
• Improved accuracy
• Easy monitoring
• Real-time reporting

### Disadvantages

• Requires internet connection
• Salesforce license limitations
• Limited offline functionality

## 9. CONCLUSION

The Farm Management System successfully digitizes agricultural data using Salesforce. It helps farmers and managers manage records efficiently, reduce errors, and analyze farm productivity through reports and dashboards.

## 10. FUTURE SCOPE

• Mobile application integration
• AI-based crop prediction
• Weather data integration
• IoT-based farm monitoring
• Market price analysis

## 11. APPENDIX

Tools Used

• Salesforce Developer Org
• Reports and Dashboards
• Custom Objects and Fields

Dataset

• Farm Records
• Crop Records
• Farmer Records
• Buyer Records